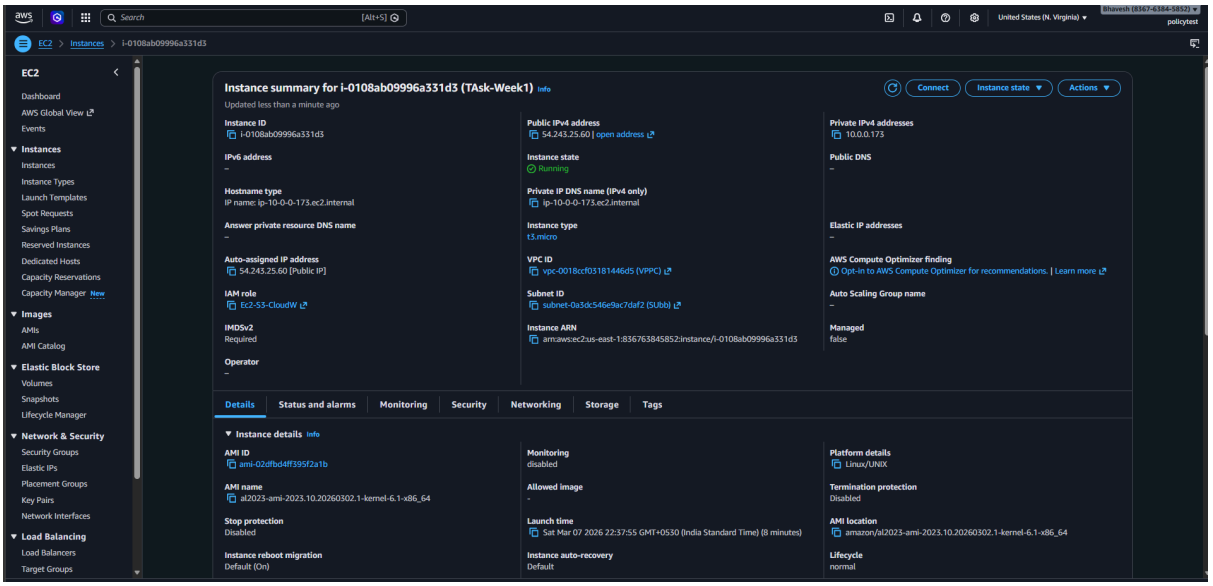
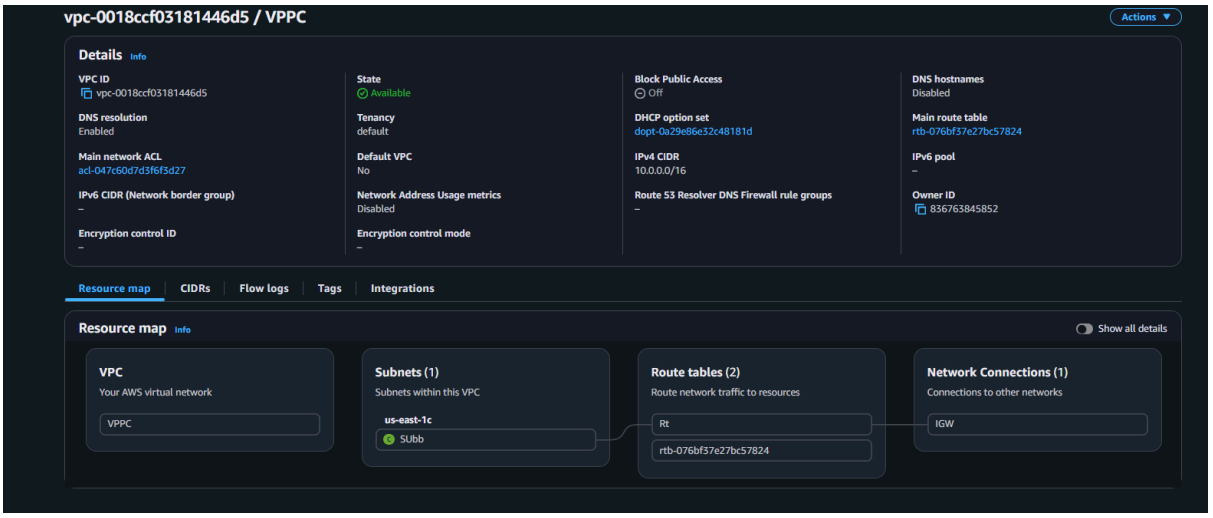
Week 1 Task

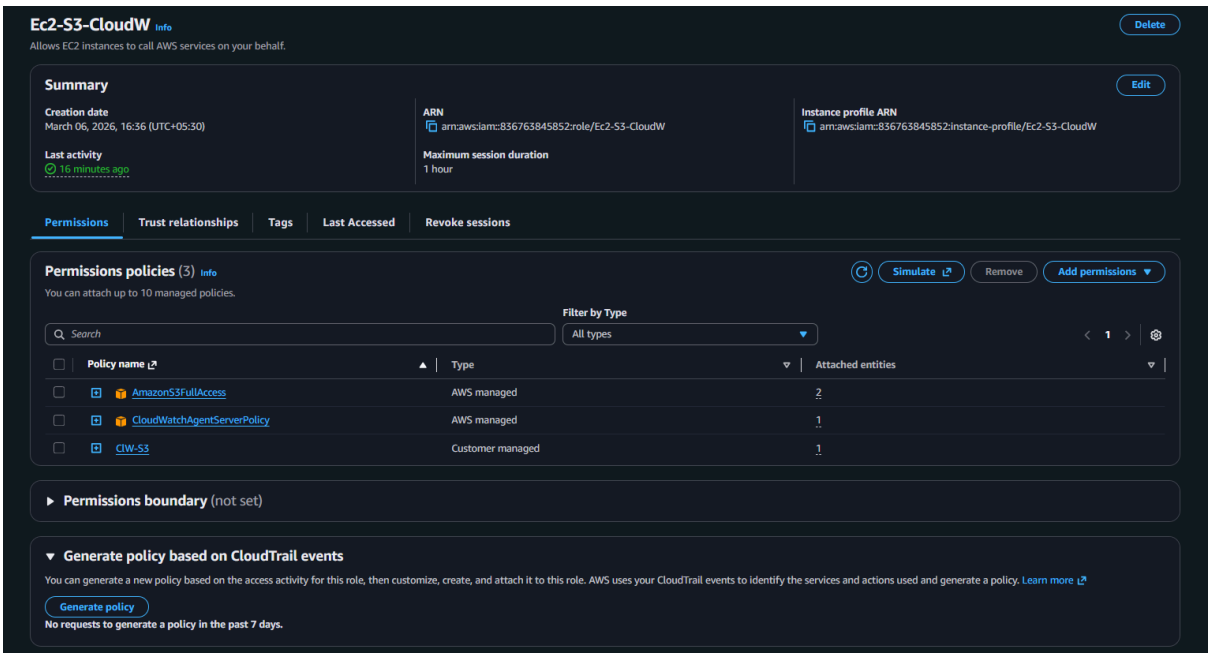
EC2

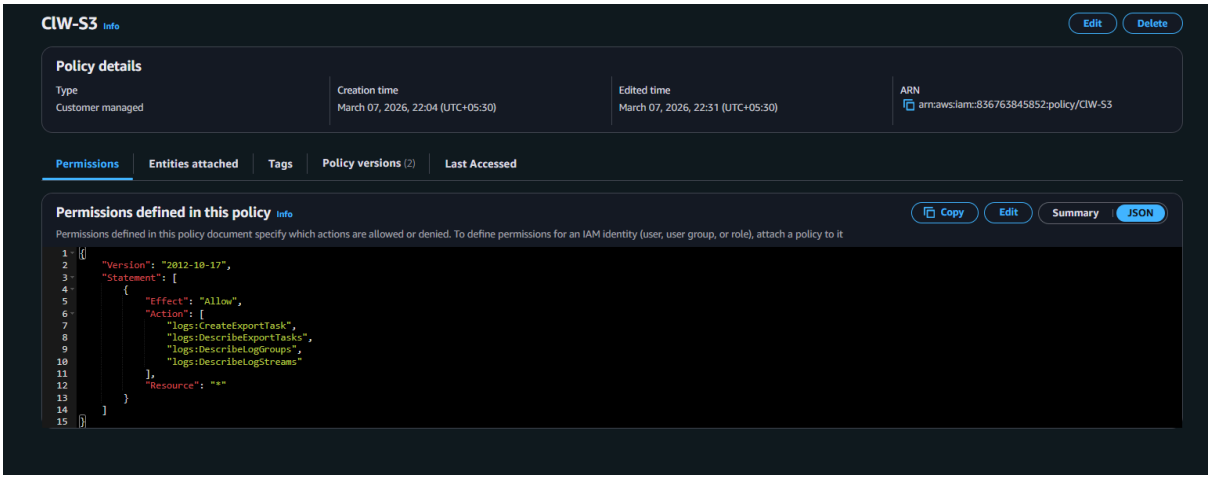


VPC

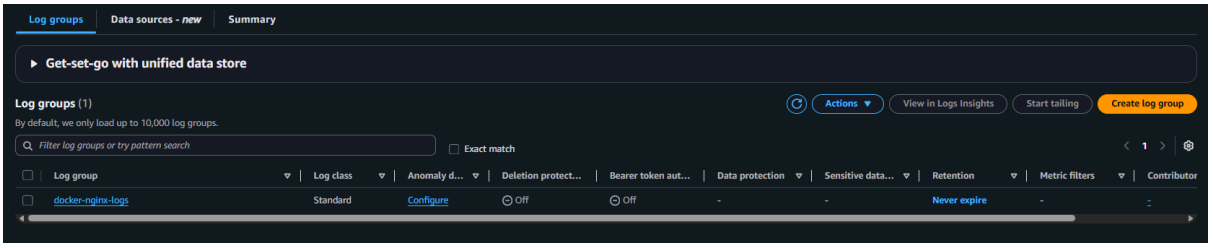


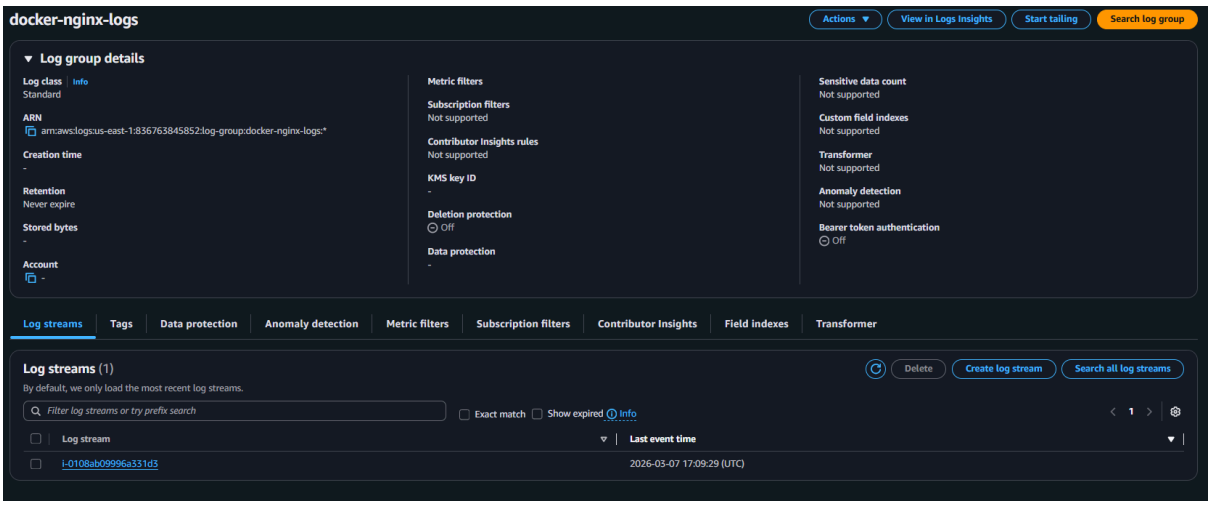
IAM

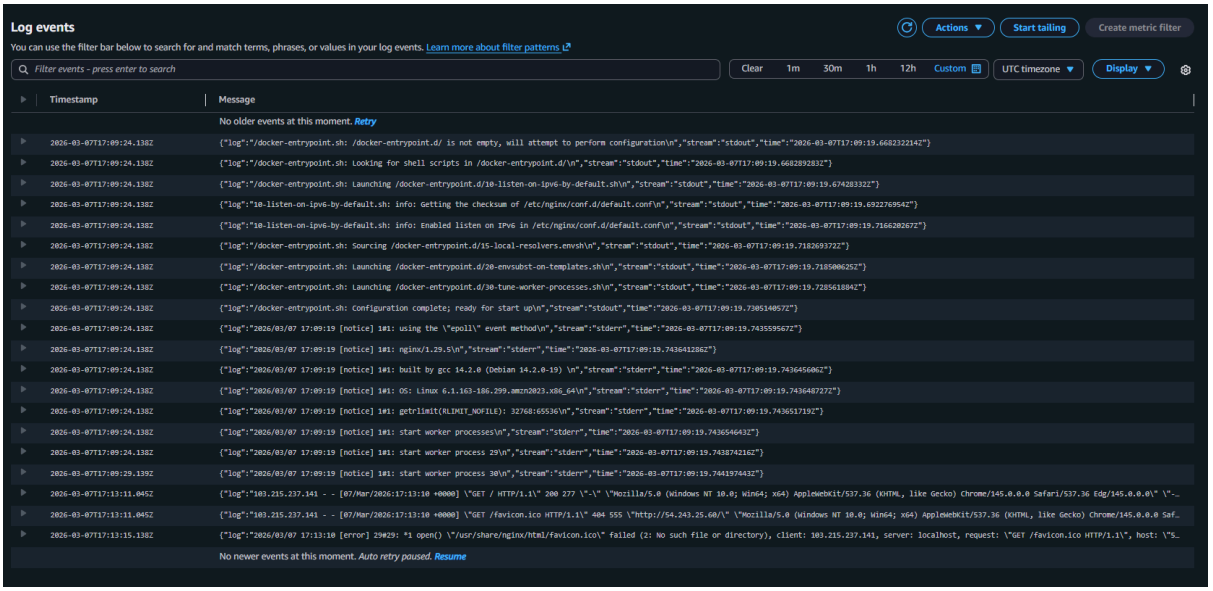

Ec2-S3-CloudW Info
Delete
Allows EC2 instances to call AWS services on your behalf.
Summary
Edit
Creation date
March 06, 2026, 16:36 (UTC+05:30)
ARN
arn:aws:iam::836763845852:role/Ec2-S3-CloudW
Instance profile ARN
arn:aws:iam::836763845852:instance-profile/Ec2-S3-CloudW
Last activity
16 minutes ago
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies (3) Info
Simulate
Remove
Add permissions
You can attach up to 10 managed policies.
Filter by Type
Search
All types
Policy name
Type
Attached entities
AmazonS3FullAccess
AWS managed
2
CloudWatchAgentServerPolicy
AWS managed
1
CIW-S3
Customer managed
1
Permissions boundary (not set)
Generate policy based on CloudTrail events
You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more
Generate policy
No requests to generate a policy in the past 7 days.


CIW-S3 Info
Edit
Delete
Policy details
Type
Customer managed
Creation time
March 07, 2026, 22:04 (UTC+05:30)
Edited time
March 07, 2026, 22:31 (UTC+05:30)
ARN
arn:aws:iam::836763845852:policy/CIW-S3
Permissions
Entities attached
Tags
Policy versions (2)
Last Accessed
Permissions defined in this policy Info
Copy
Edit
Summary
JSON
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
"logs:CreateExportTask",
"logs:DescribeExportTasks",
"logs:DescribeLogGroups",
"logs:DescribeLogStreams"
],
"Resource": "*"
}
]
}

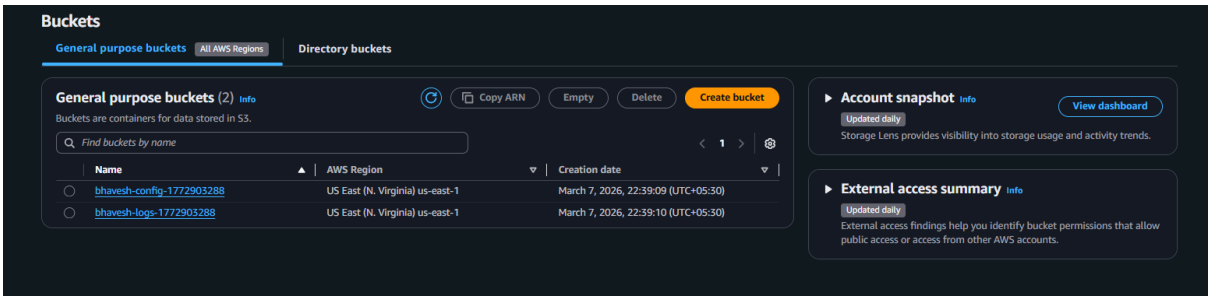
CloudWatch Logs

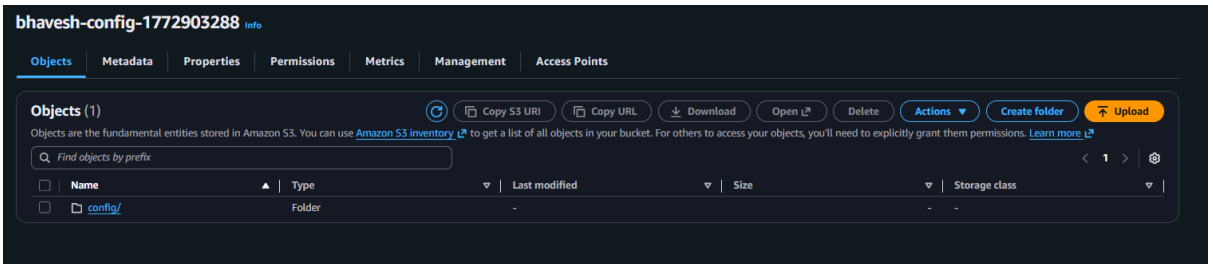


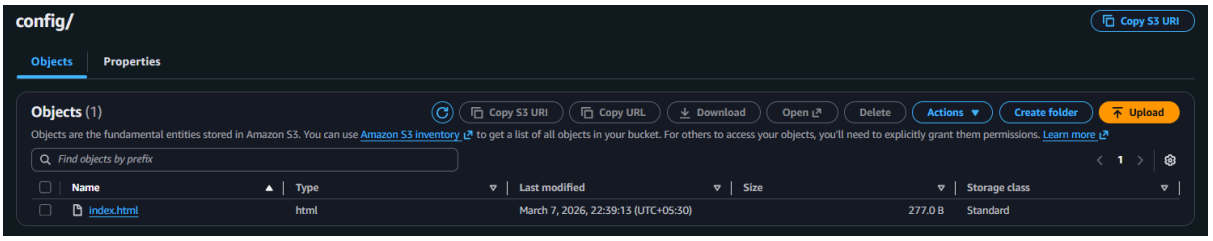


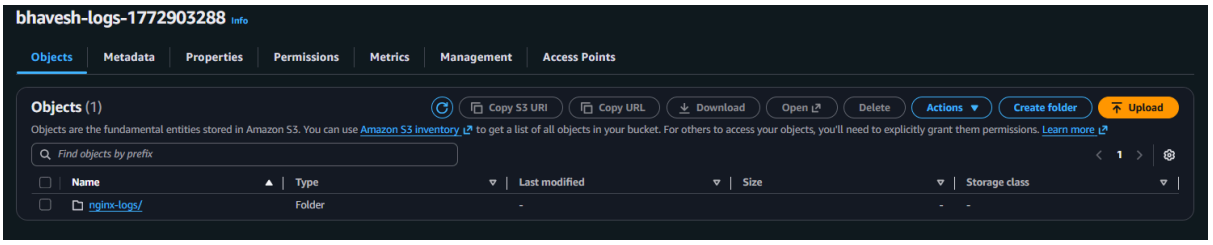


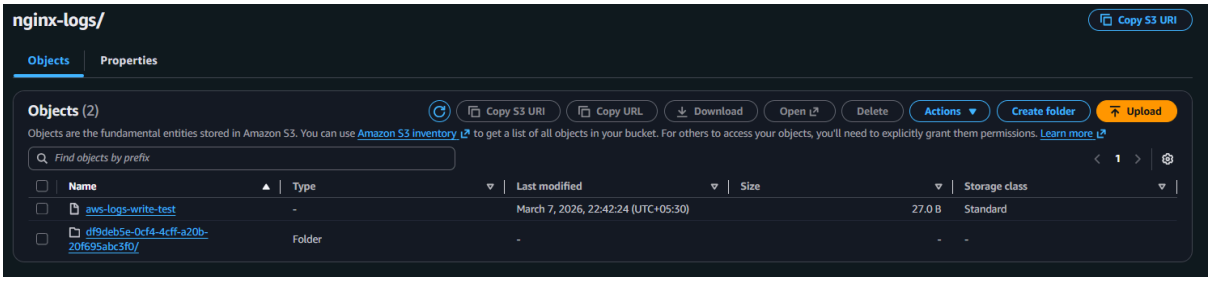
S3

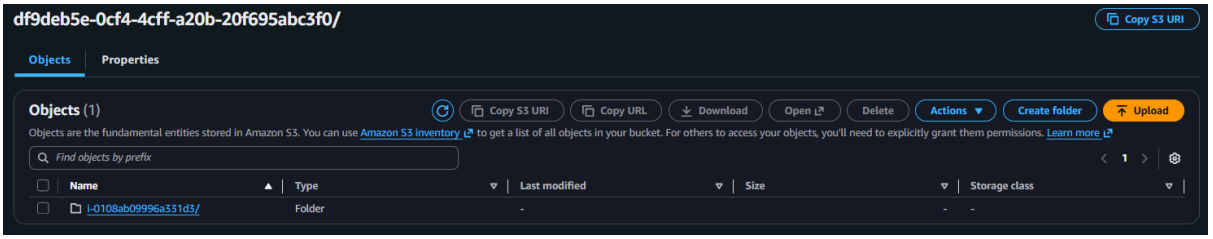

Buckets
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (2) Info
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
Name
AWS Region
Creation date
bhavesh-config-1772903288
US East (N. Virginia) us-east-1
March 7, 2026, 22:39:09 (UTC+05:30)
bhavesh-logs-1772903288
US East (N. Virginia) us-east-1
March 7, 2026, 22:39:10 (UTC+05:30)
Account snapshot Info
View dashboard
Updated daily
Storage Lens provides visibility into storage usage and activity trends.
External access summary Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.


bhavesh-config-1772903288 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
config/
Folder

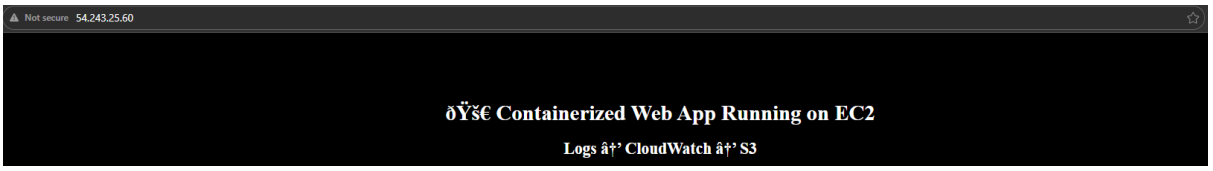

config/
Copy S3 URI
Objects
Properties
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
index.html
html
March 7, 2026, 22:39:13 (UTC+05:30)
277.0 B
Standard


bhavesh-logs-1772903288 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
nginx-logs/
Folder


nginx-logs/
Copy S3 URI
Objects
Properties
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
aws-logs-write-test
March 7, 2026, 22:42:24 (UTC+05:30)
27.0 B
Standard
df9deb5e-0cf4-4cff-a20b-20f695abc3f0/
Folder



Public IP



Script

```
#!/bin/bash
exec > /var/log/user-data.log 2>&1
set -e

REGION=us-east-1
CONFIG_BUCKET=bhavesh-config-$(date +%s)
LOG_BUCKET=bhavesh-logs-$(date +%s)

##############################################
# Update system
##############################################

dnf update -y

##############################################
# Install packages
##############################################

dnf install -y docker awscli amazon-cloudwatch-agent

##############################################
# Start Docker
```

```
##############################################

systemctl enable docker
systemctl start docker

usermod -aG docker ec2-user

##############################################
# Create S3 buckets
##############################################

aws s3api create-bucket --bucket $CONFIG_BUCKET --region $REGION
aws s3api create-bucket --bucket $LOG_BUCKET --region $REGION

##############################################
# Bucket policy for CloudWatch export
##############################################

cat <<EOF > /tmp/s3-log-policy.json
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Effect": "Allow",
   "Principal": {
    "Service": "logs.$REGION.amazonaws.com"
   },
   "Action": [
    "s3:PutObject",
    "s3:GetBucketAcl"
   ],
   "Resource": [
     "arn:aws:s3:::$LOG_BUCKET",
     "arn:aws:s3:::$LOG_BUCKET/*"
   ]
  }
 ]
}
EOF

aws s3api put-bucket-policy \
--bucket $LOG_BUCKET \
--policy file:///tmp/s3-log-policy.json

##############################################
# Create custom HTML
##############################################
```

```
cat <<EOF > /tmp/index.html
<html>
<head>
<title>Bhavesh DevOps Project</title>
</head>
<body style="background:black;color:white;text-align:center;margin-top:100px;">
<h1>🚀 Containerized Web App Running on EC2</h1>
<h2>Logs → CloudWatch → S3</h2>
<p>DevOps Monitoring Pipeline</p>
</body>
</html>
EOF

##############################################
# Upload config to S3
##############################################

aws s3 cp /tmp/index.html s3://$CONFIG_BUCKET/config/index.html

##############################################
# Pull config from S3
##############################################

mkdir -p /home/ec2-user/nginx

aws s3 cp \
s3://$CONFIG_BUCKET/config/index.html \
/home/ec2-user/nginx/index.html

##############################################
# Run Nginx container
##############################################

docker pull nginx

docker run -d \
--name nginx-webapp \
--restart always \
-p 80:80 \
-v /home/ec2-user/nginx/index.html:/usr/share/nginx/html/index.html \
nginx

##############################################
# Create CloudWatch log group
##############################################

aws logs create-log-group \
--log-group-name docker-nginx-logs \
```

```
--region $REGION || true

##############################################
# CloudWatch Agent config
##############################################

cat <<EOF > /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json
{
 "logs": {
  "logs_collected": {
   "files": {
    "collect_list": [
     {
      "file_path": "/var/lib/docker/containers/*/*.log",
      "log_group_name": "docker-nginx-logs",
      "log_stream_name": "{instance_id}"
     }
    ]
   }
  }
 }
}
EOF

##############################################
# Start CloudWatch Agent
##############################################

/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json \
-s

##############################################
# Wait for logs to generate
##############################################

sleep 180

##############################################
# Export logs to S3
##############################################

aws logs create-export-task \
--task-name nginx-logs-to-s3 \
--log-group-name docker-nginx-logs \
--from 0 \
```

```
--to $(date +%s)000 \
--destination $LOG_BUCKET \
--destination-prefix nginx-logs || true

##############################################
# Lifecycle rule for log bucket
##############################################

cat <<EOF > /tmp/lifecycle.json
{
 "Rules": [
  {
   "ID": "DeleteLogsAfter90Days",
   "Status": "Enabled",
   "Filter": {},
   "Expiration": { "Days": 90 }
  }
 ]
}
EOF

aws s3api put-bucket-lifecycle-configuration \
--bucket $LOG_BUCKET \
--lifecycle-configuration file:///tmp/lifecycle.json

echo "Setup Complete: EC2 → Docker → CloudWatch → S3"
```